

Lecture 13: Quantum mechanics and molecular structure-4

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Classroom: Teaching Center 401

Wednesday/Friday: 10:15-11:00; 11:10-11:55

2025/11/26

Major points of last lecture

- **Quantum mechanics and molecular structure**

- 8. Valence Bond Theory and the Electron Pair Bond

- 9. Orbital Hybridization for Polyatomic Molecules

- 10. Predicting Molecular Structures and Shapes

- 11. Using the LCAO and Valence Bond Methods Together

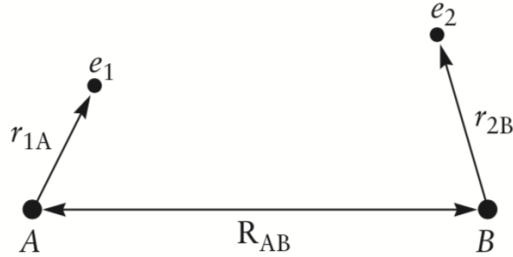
- 12. Summary and Comparison of the LCAO and Valence Bond Methods

Valence Bond Theory and the Electron Pair Bond

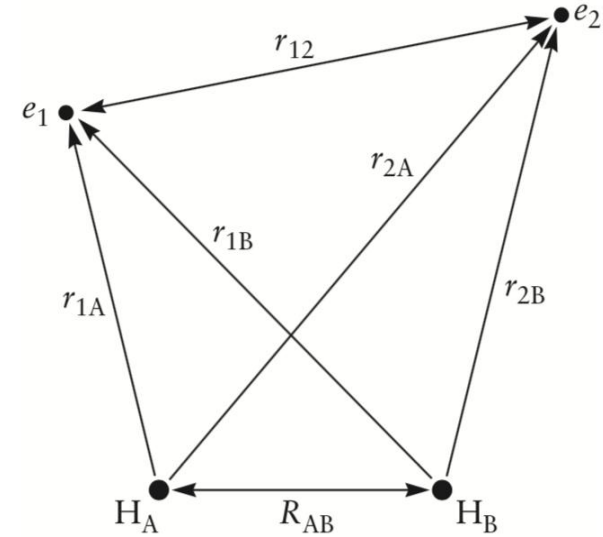
Valence bond (VB) model:

- Grew out of the **qualitative Lewis electron-pair model** in which the chemical bond is described as **a pair of electrons that is localized between two atoms**.
- The VB model **constructs a wave function for the bond** by assuming that **each atom arrives with at least one unpaired electron in an AO**.
- The resulting wave function: **a product of two one-electron wave functions**, each describing an electron localized on one of the arriving atoms. (The spins of the electrons must be paired to satisfy the Pauli exclusion principle.)

Single Bonds (H-H)



$$\varphi^A(r_{1A})\varphi^B(r_{2B})$$



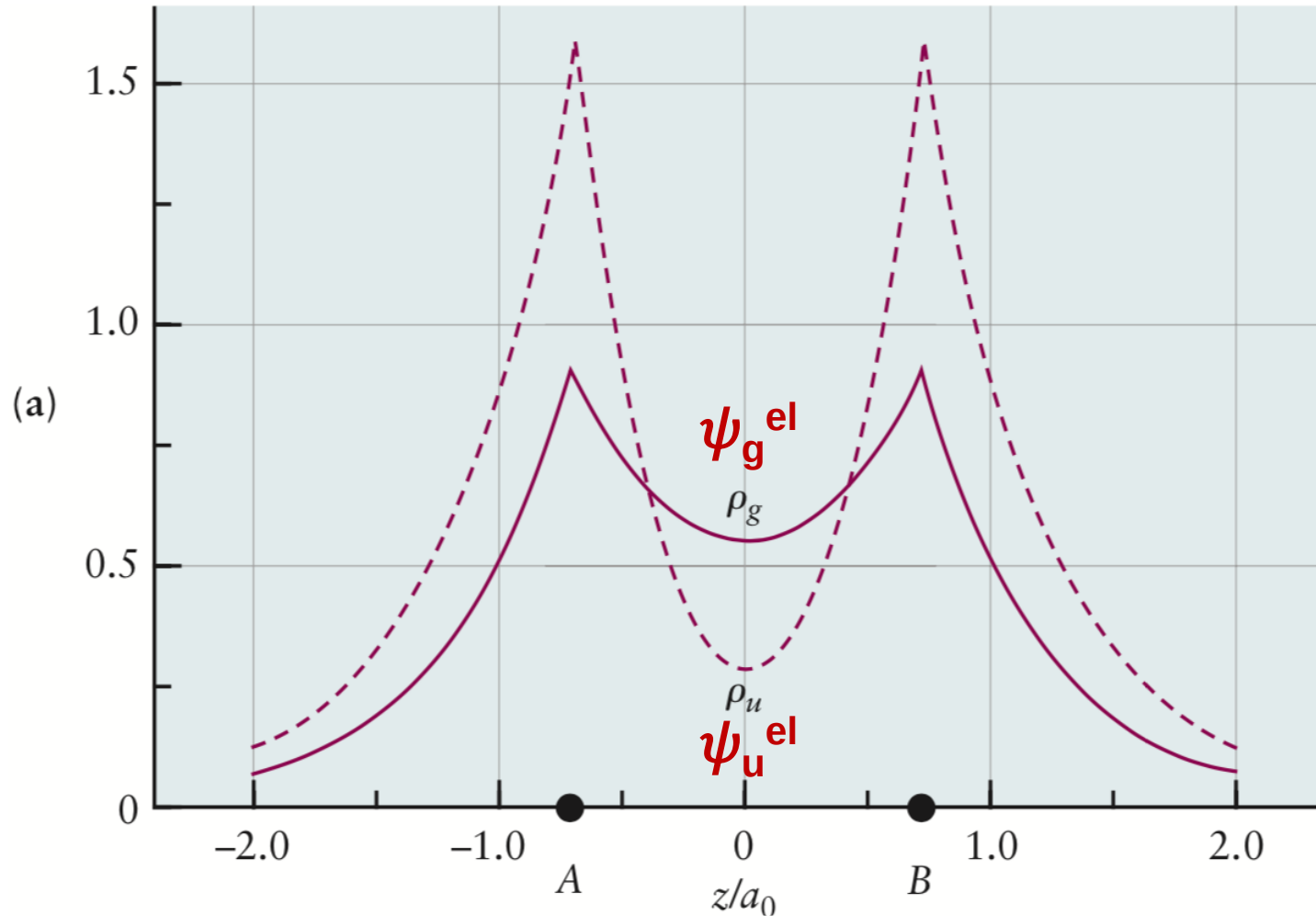
$$\psi^{\text{el}}(r_{1A}, r_{2B}; R_{AB}) = c(R_{AB})\varphi^A(r_{1A})\varphi^B(r_{2B})$$

The electrons are indistinguishable: as the atoms begin to interact strongly, we cannot determine whether electron 1 arrived with proton A and electron 2 with proton B, or vice versa.

$$\psi^{\text{el}}(r_{1A}, r_{2B}; R_{AB}) = c_1(R_{AB})\varphi^A(r_{1A})\varphi^B(r_{2B}) + c_2(R_{AB})\varphi^A(r_{2A})\varphi^B(r_{1B}) \quad c_1 = c_2 \text{ and } c_1 = -c_2$$

Symmetry requires that $c_1 = c_2$ and $c_1 = -c_2$ are equally valid choices

Single Bonds (H-H)



$$\psi_g^{\text{el}} = C_1[1s^{\text{A}}(1)1s^{\text{B}}(2) + 1s^{\text{A}}(2)1s^{\text{B}}(1)]$$



FIGURE 6.24 The electron density for the ψ_g^{el} and ψ_u^{el} wave functions in the simple valence bond model for H_2 . (a) The electron density ρ_g for ψ_g^{el} and ρ_u for ψ_u^{el} calculated analytically as described in the text. (b) Three-dimensional isosurface of the electron density for the ψ_g^{el} wave function obtained by computer calculations.

(Courtesy of Mr. Hatem Helal and Professor William A. Goddard III, California Institute of Technology, and Dr. Kelly P. Gaither, University of Texas at Austin.)

Characteristics of the two theories

LCAO

- Ignores e-e repulsion
- Electrons wander around
- Orbitals: delocalized (离域)

VB

- Exaggerates e-e repulsion
- Electrons adhere to the nuclei
- Orbitals: localized (定域/局域)

Single Bonds (F-F)



Lewis diagram for F₂.

$$\psi_g^{\text{bond}} = C_1[2p_z^{\text{A}}(1)2p_z^{\text{B}}(2) + 2p_z^{\text{A}}(2)2p_z^{\text{B}}(1)]$$

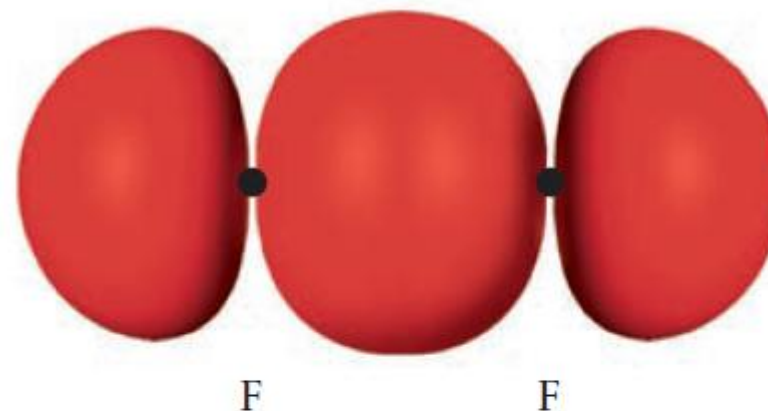


FIGURE 6.25 Isosurface representation of the electron density in the F₂ σ bond formed from a pair of electrons initially localized in a 2p_z orbital on each F atom. (Courtesy of Mr. Hatem Helal and Professor William A. Goddard III, California Institute of Technology, and Dr. Kelly P. Gaither, University of Texas at Austin.)

Single Bonds (H-F)

HF: one shared pair in a single bond produced by overlap of H(1s) and F(2p_z).

$$\psi^{\text{bond}} = C_1[2p_z^{\text{F}}(1)1s^{\text{H}}(2)] + C_2[2p_z^{\text{F}}(2)1s^{\text{H}}(1)]$$

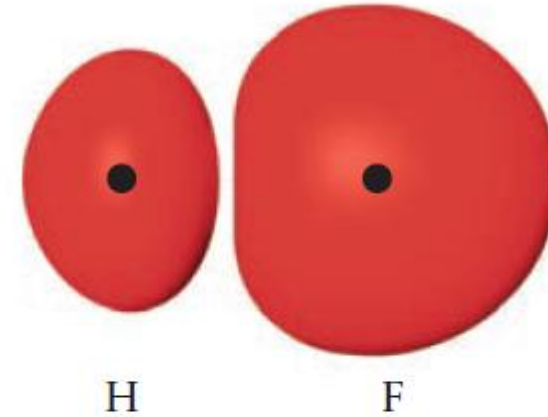
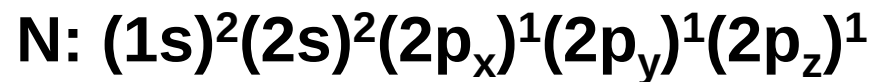


FIGURE 6.26 Isosurface representation of the electron density in the HF σ bond formed from a pair of electrons initially localized in a 1s orbital on H and in a 2p_z orbital on F. (Courtesy of Mr. Hatem Helal and Professor William A. Goddard III, California Institute of Technology, and Dr. Kelly P. Gaither, University of Texas at Austin.)

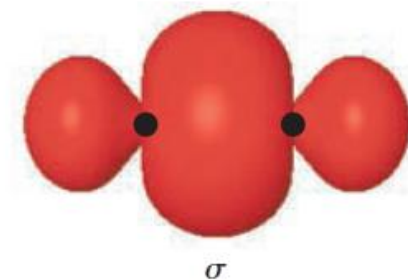
Multiple Bonds (N₂)



Lewis diagram for N₂.

- When approach one another along the z-axis (2p_z: head-on)

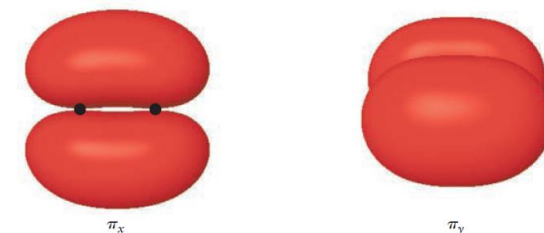
$$\psi_{\sigma}^{\text{bond}} = C_1[2p_z^{\text{A}}(1)2p_z^{\text{B}}(2) + 2p_z^{\text{A}}(2)2p_z^{\text{B}}(1)]$$



- The 2p_x and 2p_y orbitals: side-to-side.

$$\psi_{\pi_x}^{\text{bond}}(1, 2; R_{\text{AB}}) = C_1(R_{\text{AB}})[2p_x^{\text{A}}(1)2p_x^{\text{B}}(2)] + C_1(R_{\text{AB}})[2p_x^{\text{A}}(2)2p_x^{\text{B}}(1)]$$

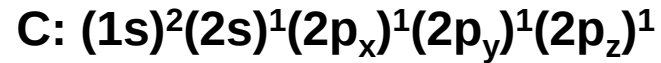
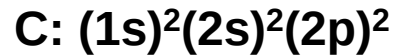
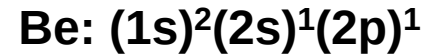
$$\psi_{\pi_y}^{\text{bond}}(1, 2) = C_1[2p_y^{\text{A}}(1)2p_y^{\text{B}}(2) + 2p_y^{\text{A}}(2)2p_y^{\text{B}}(1)]$$



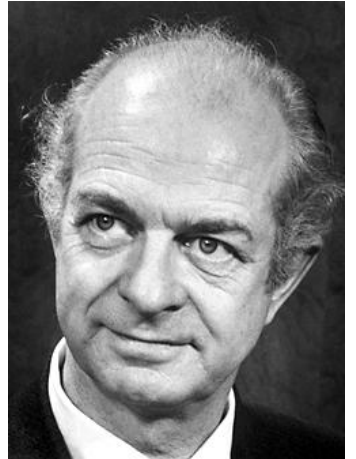
Polyatomic Molecules

Linus Pauling gave the answer in 1931: the concepts of **promotion (激发)** and **hybridization (杂化)**.

Promotion of valence electrons **from the ground state to excited states** at higher energy



Hybridization of 2s and 2p orbitals after promotion: form new orbitals with the **proper symmetry** for bond formation

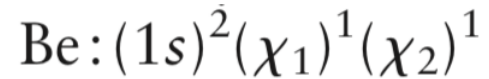


Linus Pauling
(1901–1994)

SP hybridization

BeH₂: linear

sp hybrid atomic orbitals: formed as the **sum or difference** of **one s orbital and one p orbital**.



lowercase Greek letter chi, χ , to represent hybrid orbitals

$$\chi_1(r) = \frac{1}{\sqrt{2}} [2s + 2p_z]$$

$$\chi_2(r) = \frac{1}{\sqrt{2}} [2s - 2p_z]$$

The coefficient $1/\sqrt{2}$ is a normalization constant

SP² hybridization

BH₃: a trigonal planar structure with three equivalent bonds

Promotion of one of the 2s electrons creates the excited-state configuration **B: (1s)² (2s)¹(2p_x)¹(2p_y)¹**, and these three AOs can be mixed to form the three equivalent new orbitals:

sp² hybrid atomic orbitals: formed from **one s orbital and two p orbitals**.

SP³ hybridization

CH₄: a tetrahedral structure with four equivalent bonds

Promotion of one of the 2s electrons creates the excited-state configuration **C: (1s)² (2s)¹(2p_x)¹(2p_y)¹(2p_z)¹**, and these four AOs can be mixed to form the four equivalent new orbitals:

$$\chi_1(r) = \frac{1}{2} [2s + 2p_x + 2p_y + 2p_z]$$

$$\chi_2(r) = \frac{1}{2} [2s - 2p_x - 2p_y + 2p_z]$$

$$\chi_3(r) = \frac{1}{2} [2s + 2p_x - 2p_y - 2p_z]$$

$$\chi_4(r) = \frac{1}{2} [2s - 2p_x + 2p_y - 2p_z]$$

sp³ hybrid atomic orbitals: formed from **one s orbital and three p orbitals**.

The Valence Shell Electron-Pair Repulsion Theory (VSEPR)

- The Valence Shell Electron-Pair Repulsion Theory (VSEPR): 价层电子对互斥模型



- fundamental idea: the electron pairs position themselves as far apart as possible to minimize their repulsions



$$SN = \left(\begin{array}{c} \text{number of atoms} \\ \text{bonded to central atom} \end{array} \right) + \left(\begin{array}{c} \text{number of lone pairs} \\ \text{on central atom} \end{array} \right)$$

SN: steric number (空间位数)



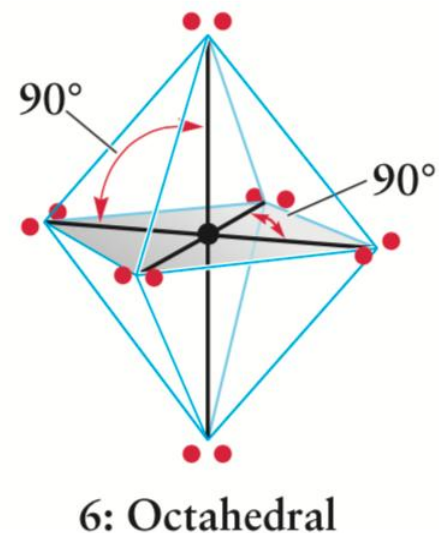
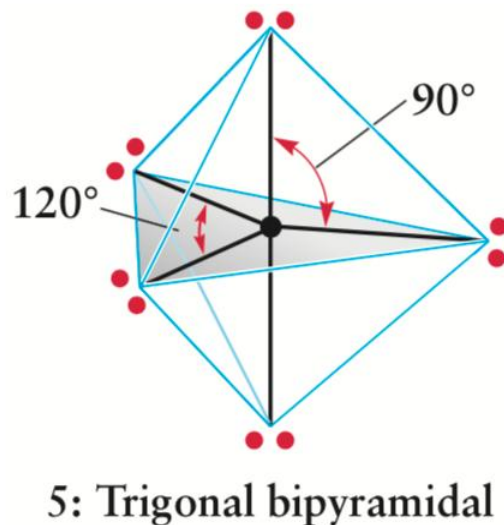
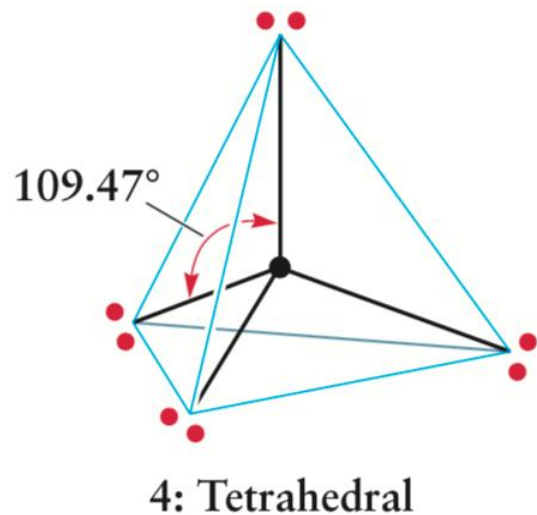
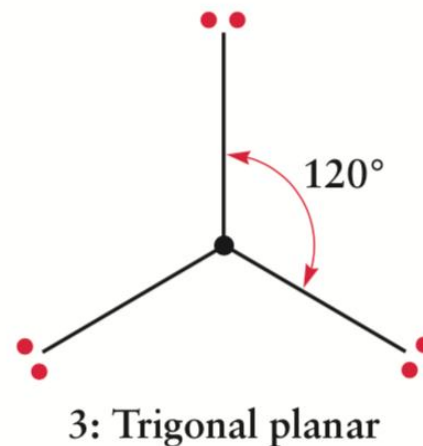
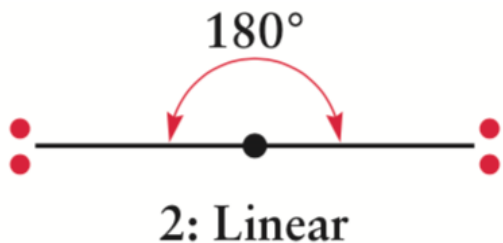
Ronald J. Gillespie
(1924–)



Ronald S. Nyholm
(1917–1971)

The Valence Shell Electron-Pair Repulsion Theory (VSEPR)

SN: steric number (空间位数)



The Valence Shell Electron-Pair Repulsion Theory (VSEPR)

Step 1: Rewrite a structure as AX_mE_n , m = number of ligands, n = number of lone pairs

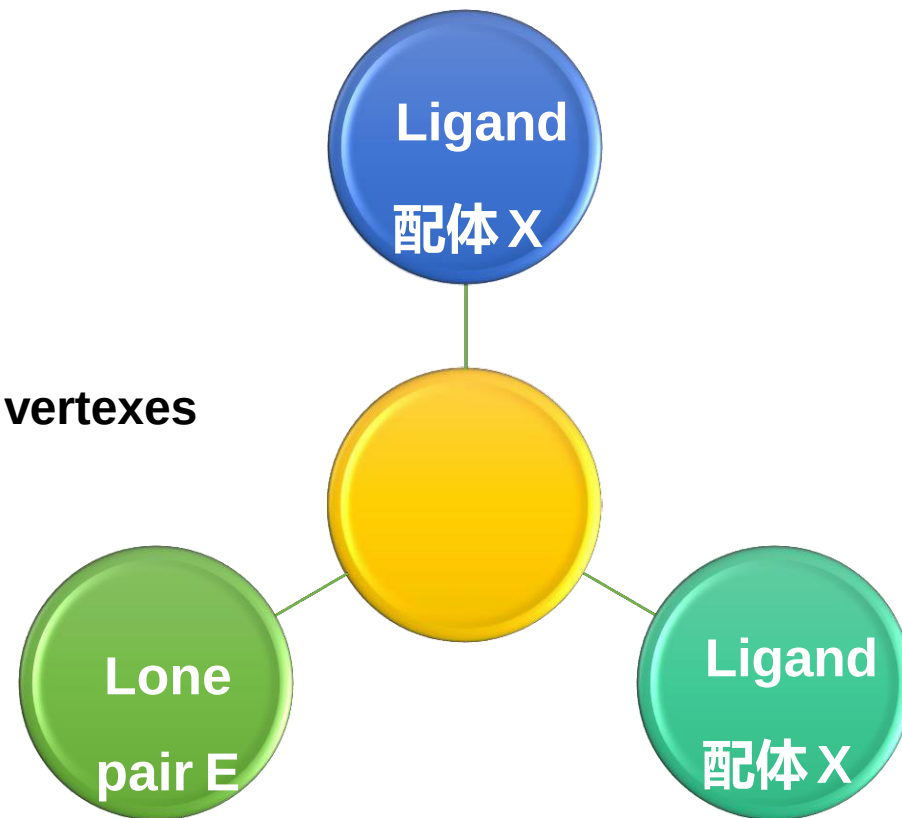
Step 2: Assign a Steric Number; $SN = m + n$

Step 3: Place the ligands and lone pairs on a polyhedron of SN vertexes

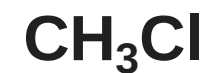
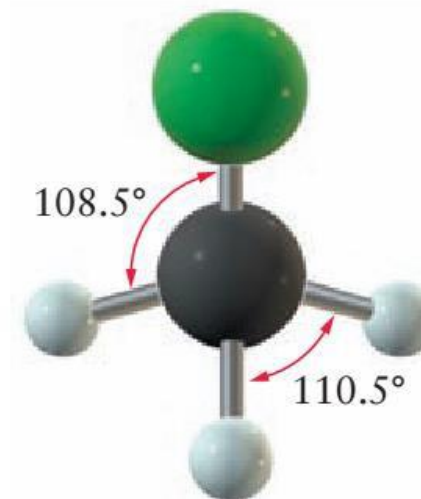
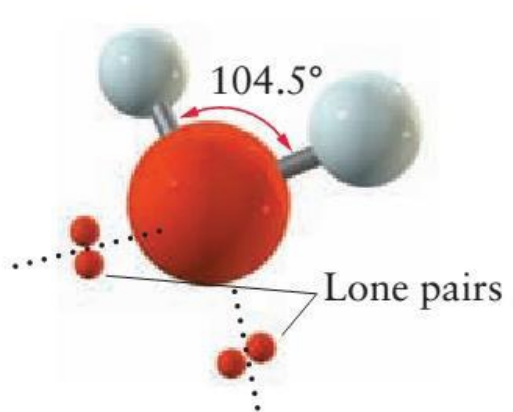
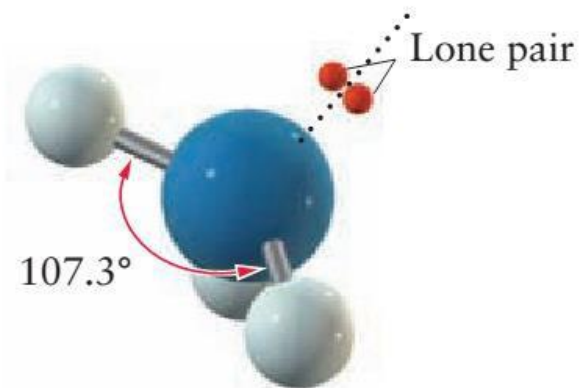
Step 4: Repulsion: $E-E > E-X > X-X$;

triple bond > double bond > single bond

X high electronegativity (EN), less repulsion



Molecular structures

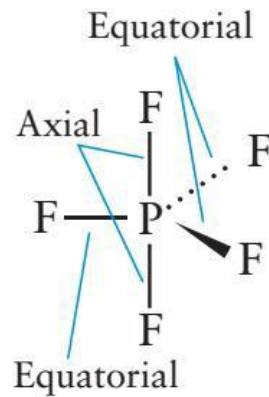


Step 4: Repulsion: $\text{E-E} > \text{E-X} > \text{X-X}$;

triple bond > double bond > single bond

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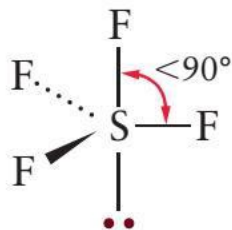
Molecular structures



(a) PF_5



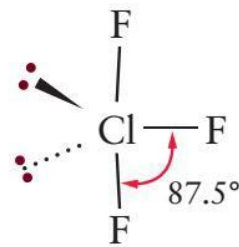
Trigonal bipyramid



(c) SF_4



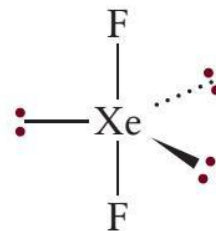
Distorted pyramid
(high-energy, not favored)



(d) ClF_3



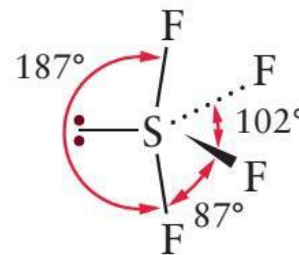
Distorted T



(e) XeF_2



Linear



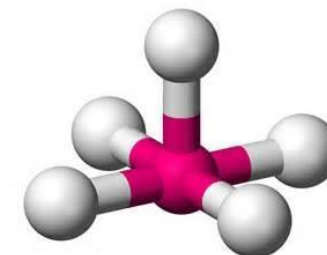
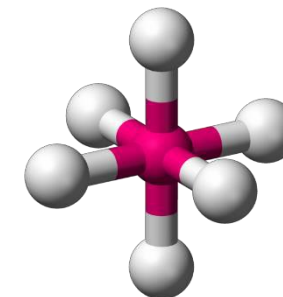
(b) SF_4



Seesaw
(low-energy, favored)

Summary of molecular geometry

SN of A	Molecular type X is a bonded atom; E is a lone pair	Predicted shape
2	AX_2	Linear (线形)
3	AX_3	Trigonal planar (三角形)
	AX_2E	Bent (弯曲形)
4	AX_4	Tetrahedral (四面体)
	AX_3E	Trigonal pyramidal (三角锥)
	AX_2E_2	Bent
5	AX_5	Trigonal bipyramid (三角双锥)
	AX_4E	Distorted see-saw (跷跷板)
	AX_3E_2	Distorted T (T字)
	AX_2E_3	Linear
6	AX_6	Octahedral (八面体)
	AX_5E	Square pyramidal (四角锥)
	AX_4E_2	Square planar (正方平面)



Major points of today's lecture

- **Quantum mechanics and molecular structure**

- 12. Summary and Comparison of the LCAO and Valence Bond Methods**

- 13. Predicting Molecular Structures and Shapes**

- 14. Using the LCAO and Valence Bond Methods Together**

- 15. Summary and Comparison of the LCAO and Valence Bond Methods**

Summary of hybridization results

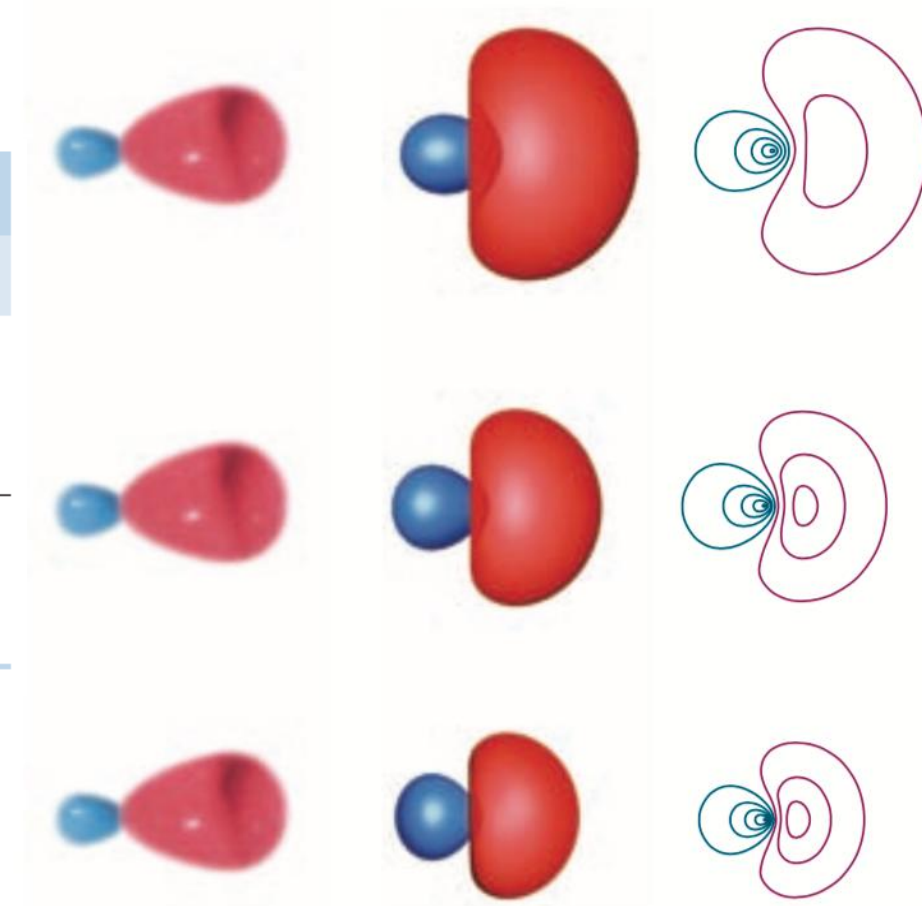
A close relationship between the **VSEPR theory** and the **hybrid orbital approach**:

- **steric numbers of 2, 3, and 4** corresponding to **sp , sp^2 , and sp^3** hybridization

TABLE 6.4

Orbital Hybridization and Molecular Geometry

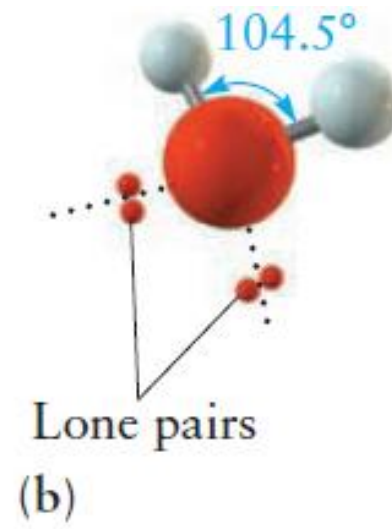
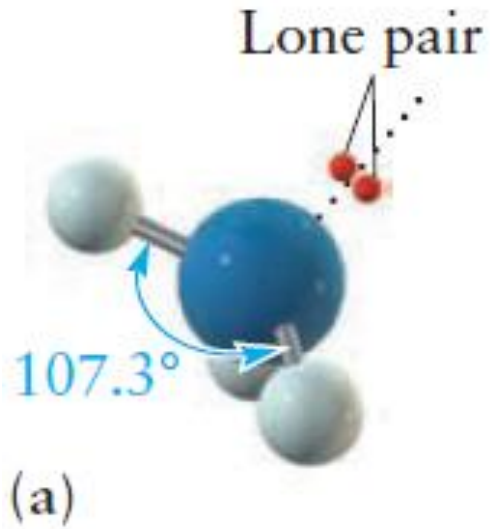
Molecule	Hybrid orbitals on central atom	Molecular geometry	Example
AX_2	sp	Linear	BeH_2
AX_3	sp^2	Trigonal planar	BH_3
AX_4	sp^3	Tetrahedral	CH_4



Hybridization and lone pairs

Lone-pair electrons, as well as **bonding pairs**, can occupy hybrid orbitals.

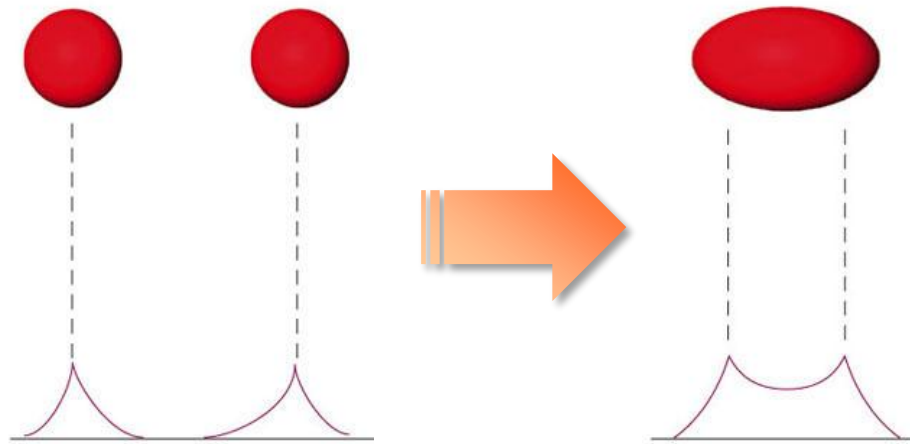
The nitrogen atom in NH_3 also has steric number 4, and its bonding can be described in terms of sp^3 hybridization.



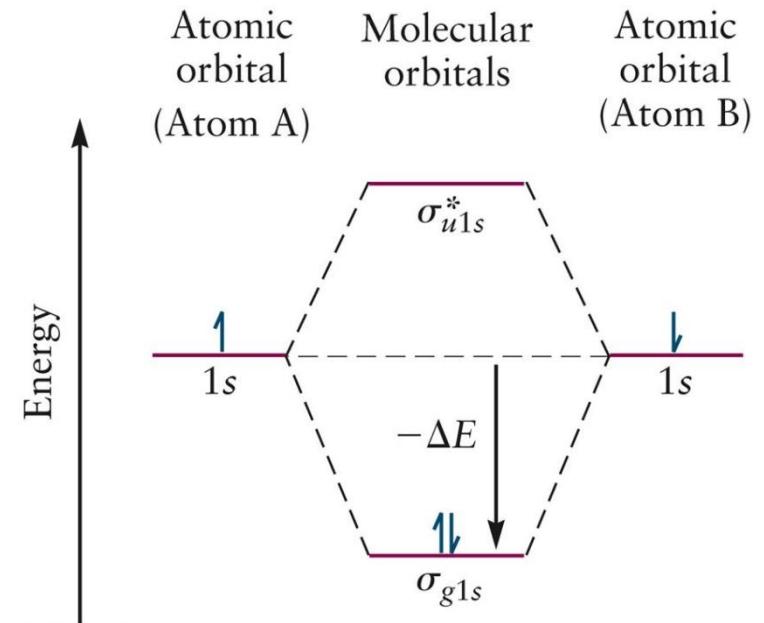
Summary: Bonding Orbital



Lewis: Shared electron pair

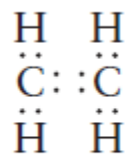


VB: Orbital overlap

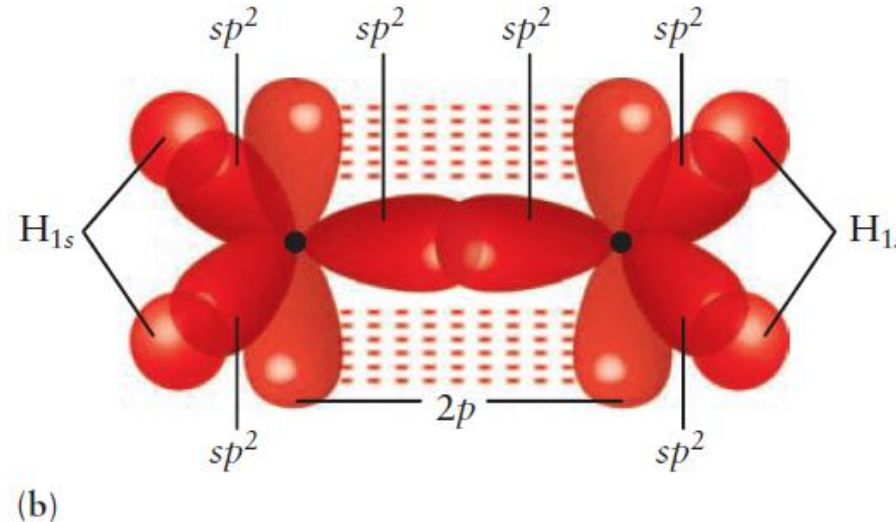
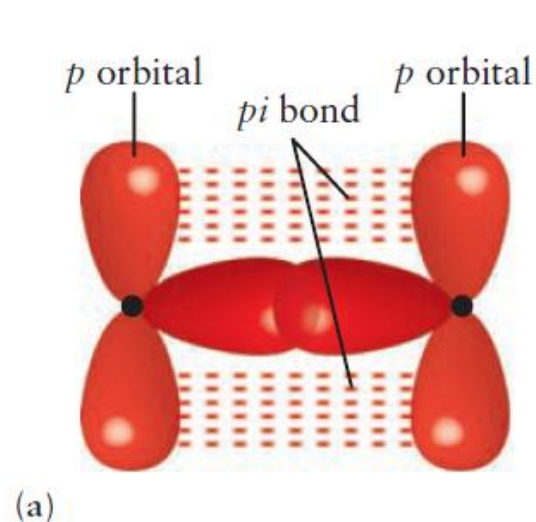
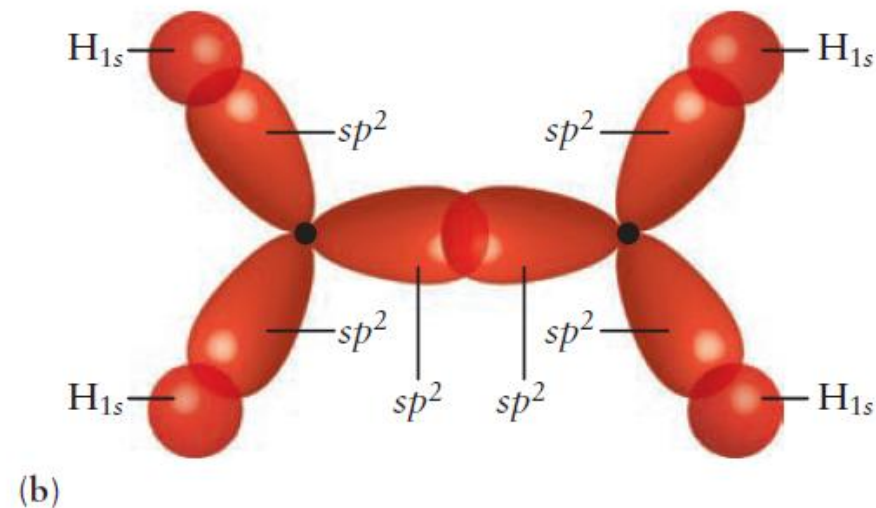
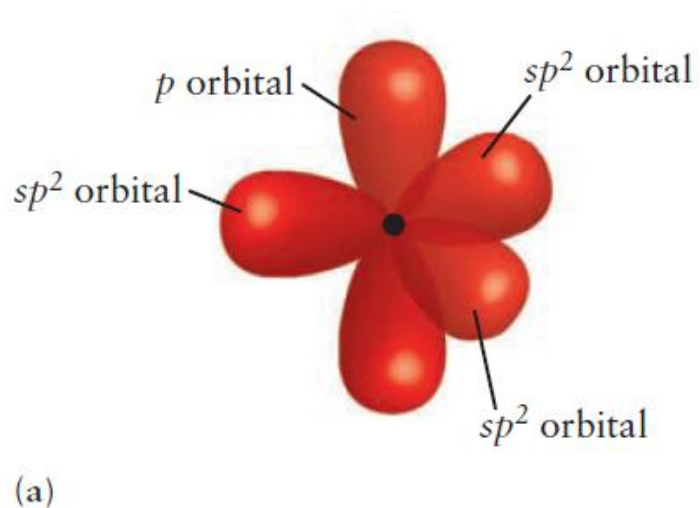


MO: Bonding orbital

Hybridization and multiple bonds in organic carbon compounds



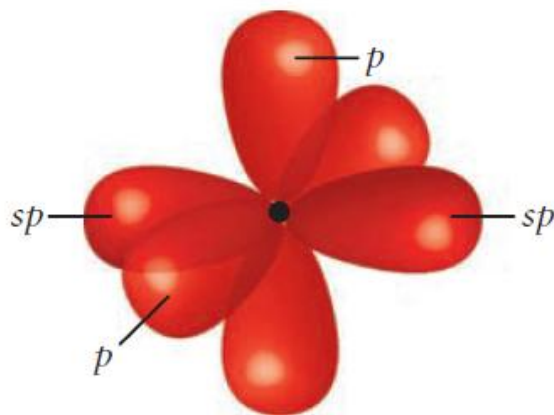
Lewis diagram for C_2H_4 .



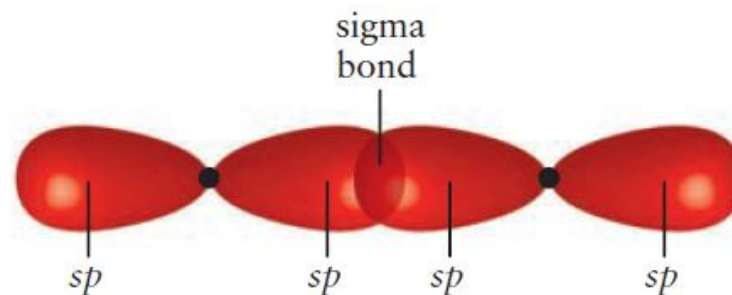
Hybridization and multiple bonds in organic carbon compounds



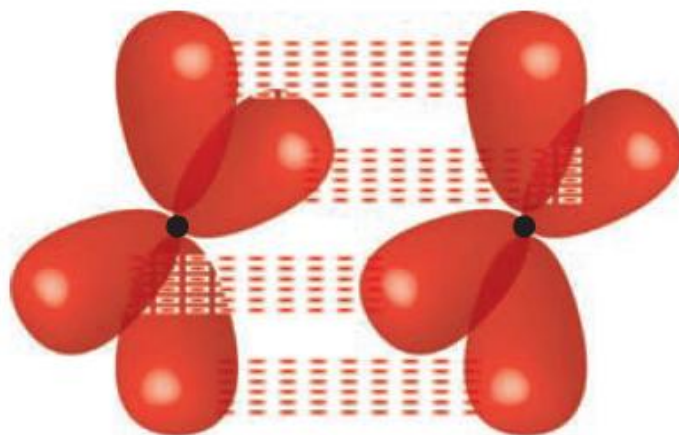
Lewis diagram for C_2H_2 .



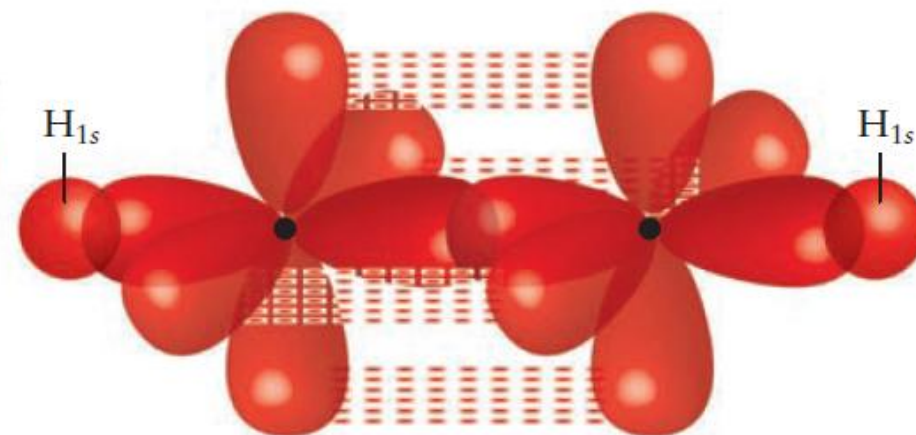
(a)



(b)

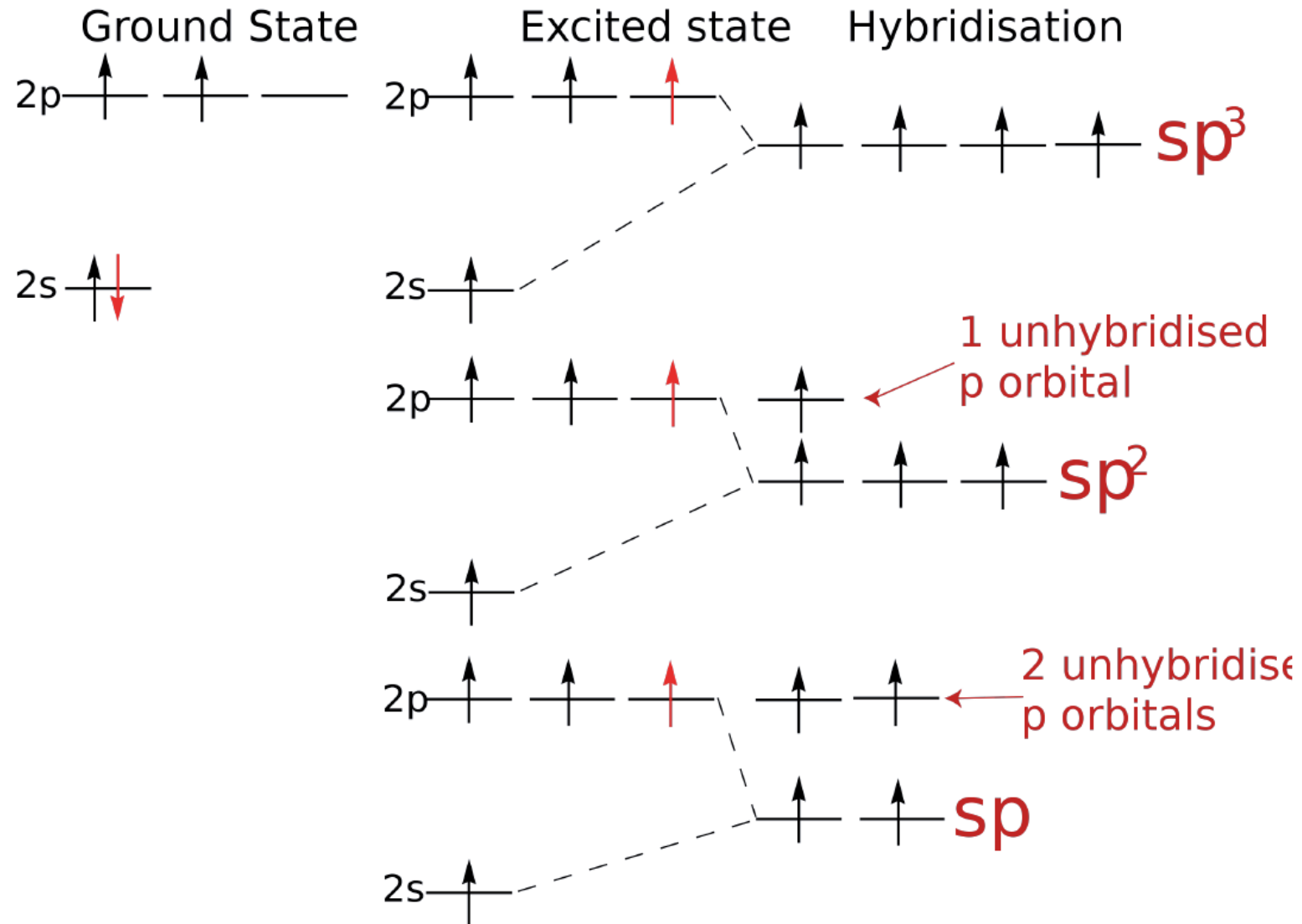


(a)



(b)

Summary of hybridization results



Major points of today's lecture

- **Quantum mechanics and molecular structure**
 - 12. Summary and Comparison of the LCAO and Valence Bond Methods
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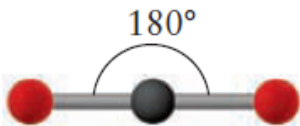
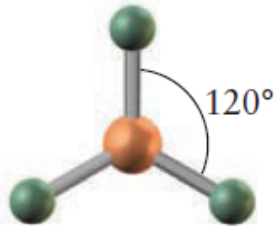
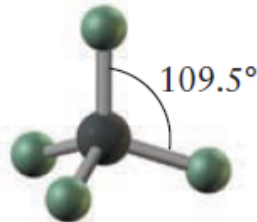
Predicting Molecular Structures and Shapes

- Determine the **empirical formula**: an **elemental analysis** based on combustion.
- Determine the **molecular formula**: its behavior as a gas or by mass spectrometry.
- Determine the **structural formula**: write a **Lewis diagram** based on the molecular formula.
- Determine the **molecular shape** experimentally, e.g., by X-ray scattering, or predict the shape by using VSEPR theory.
- Identify the **hybridization scheme** that best explains the shape predicted by VSEPR.

Predicting Molecular Structures and Shapes

TABLE 6.5

Molecular Shapes Predicted by the Valence Shell Electron-Pair Repulsion Theory and Rationalized by Orbital Hybridization

Molecule	Steric Number	Number of Lone Pairs	Orbital Hybridization	Predicted Geometry	Image	Example
AX_2	2	0	sp	Linear		BeH_2 , CO_2
AX_3	3	0	sp^2	Trigonal planar		BF_3 , SO_3
AX_2	3	1	sp^2	Bent		SO_2
AX_4	4	0	sp^3	Tetrahedral		CF_4 , SO_4^{2-}
AX_3	4	1	sp^3	Trigonal pyramidal		NH_3 , PF_3 , $AsCl_3$
AX_2	4	2	sp^3	Bent		H_2O , H_2S , SF_2

Hydrazine is a potentially explosive gas used in rocket fuels. Elemental analysis shows its mass per cent composition to be 87.419% nitrogen and 12.581% hydrogen. The density of hydrazine at 1 atm pressure and 25 °C is 1.31 g L⁻¹. Determine the molecular formula for hydrazine. Predict the structure of hydrazine by writing down its Lewis diagram and using VSEPR theory. What is the hybridization of the nitrogen atoms?

Solution

The background for the first part of this example is described in Section 2.3 and Examples 2.3 and 2.4. The number of moles of nitrogen and hydrogen in a sample of 100.00g of hydrazine are

$$\text{nitrogen} : \frac{87.42 \text{ g N}}{14.9967 \text{ g mol}^{-1}} = 6.2413 \text{ mol N}$$

$$\text{hydrogen} : \frac{12.58 \text{ g H}}{1.0079 \text{ g mol}^{-1}} = 12.4814 \text{ mol H}$$

$$\text{ratio} : \frac{\text{moles H}}{\text{moles N}} = \frac{12.4814}{6.2413} = 1.9998$$

The empirical formula is therefore close to NH₂. To determine the true molecular formula we estimate the molar mass of hydrazine using the ideal gas law (see Sec. 9.5). If m is the mass of the sample in grams and M is the molar mass in grams mol⁻¹, the molar mass M is determined from the density ρ and the gas properties as follows:

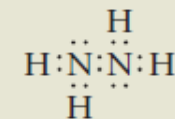
$$PV = nRT = \frac{m}{M} RT$$

$$M = \frac{mRT}{PV} = \left(\frac{m}{V} \right) \frac{RT}{P} = \rho \frac{RT}{P}$$

$$M = \frac{(1.31 \text{ g L}^{-1})(0.0821 \text{ L atm mol}^{-1} \text{ K}^{-1})(298.15 \text{ K})}{1 \text{ atm}}$$

$$M = 32.066 \text{ g mol}^{-1}$$

The molar mass corresponding to the empirical formula NH₂ is 16.0125 g mol⁻¹. Because the true molar mass is approximately twice this value the molecular formula for hydrazine is N₂H₄, which is frequently written as H₂NNH₂. The Lewis diagram is



Lewis diagram for N₂NNH₂.

Both nitrogen atoms have steric number 4 and are sp^3 hybridized, with H—N—H and H—N—N angles of approximately 109.5°. The extent of rotation about the N—N bond cannot be predicted from the VSEPR theory or the hybrid orbital model. Figure 6.37 shows the full three-dimensional structure of hydrazine.

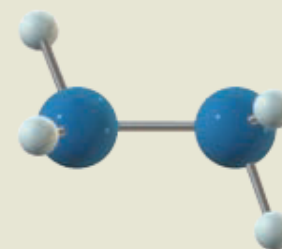
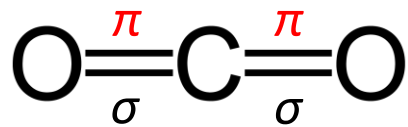
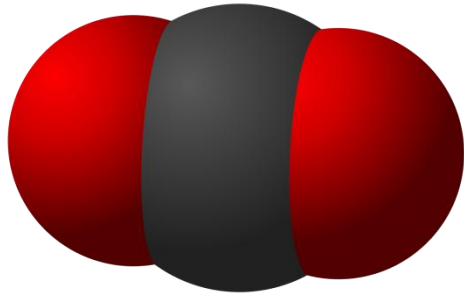


FIGURE 6.37 The structure of hydrazine, N₂H₄.

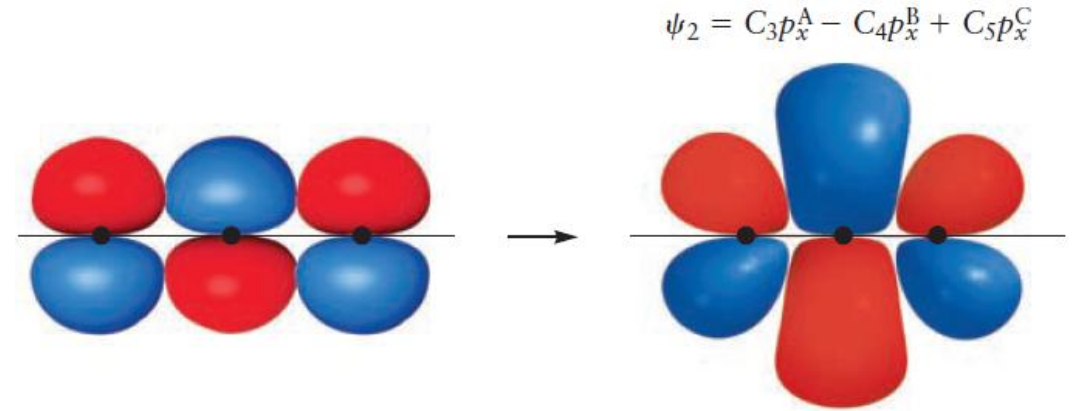
Major points of today's lecture

- **Quantum mechanics and molecular structure**
 - 12. Summary and Comparison of the LCAO and Valence Bond Methods
 - 13. Predicting Molecular Structures and Shapes
 - 14. Using the LCAO and Valence Bond Methods Together
 - 15. Summary and Comparison of the LCAO and Valence Bond Methods

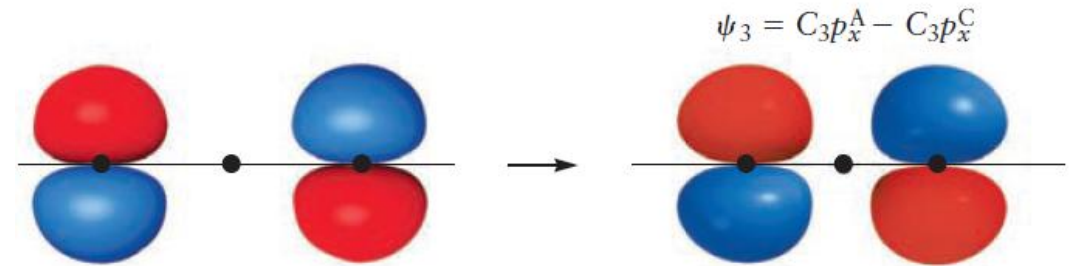
Linear Triatomic Nonhydrides



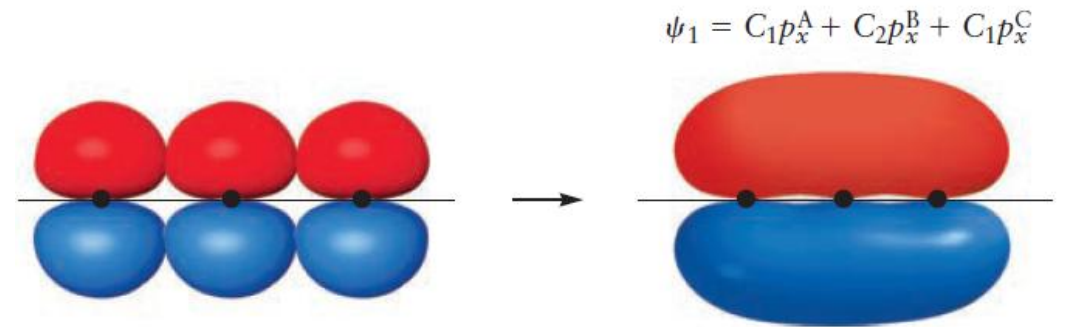
$$E_n = \frac{n^2 h^2}{8mL^2} \quad n = 1, 2, 3, \dots$$



(c) π_x^* antibonding

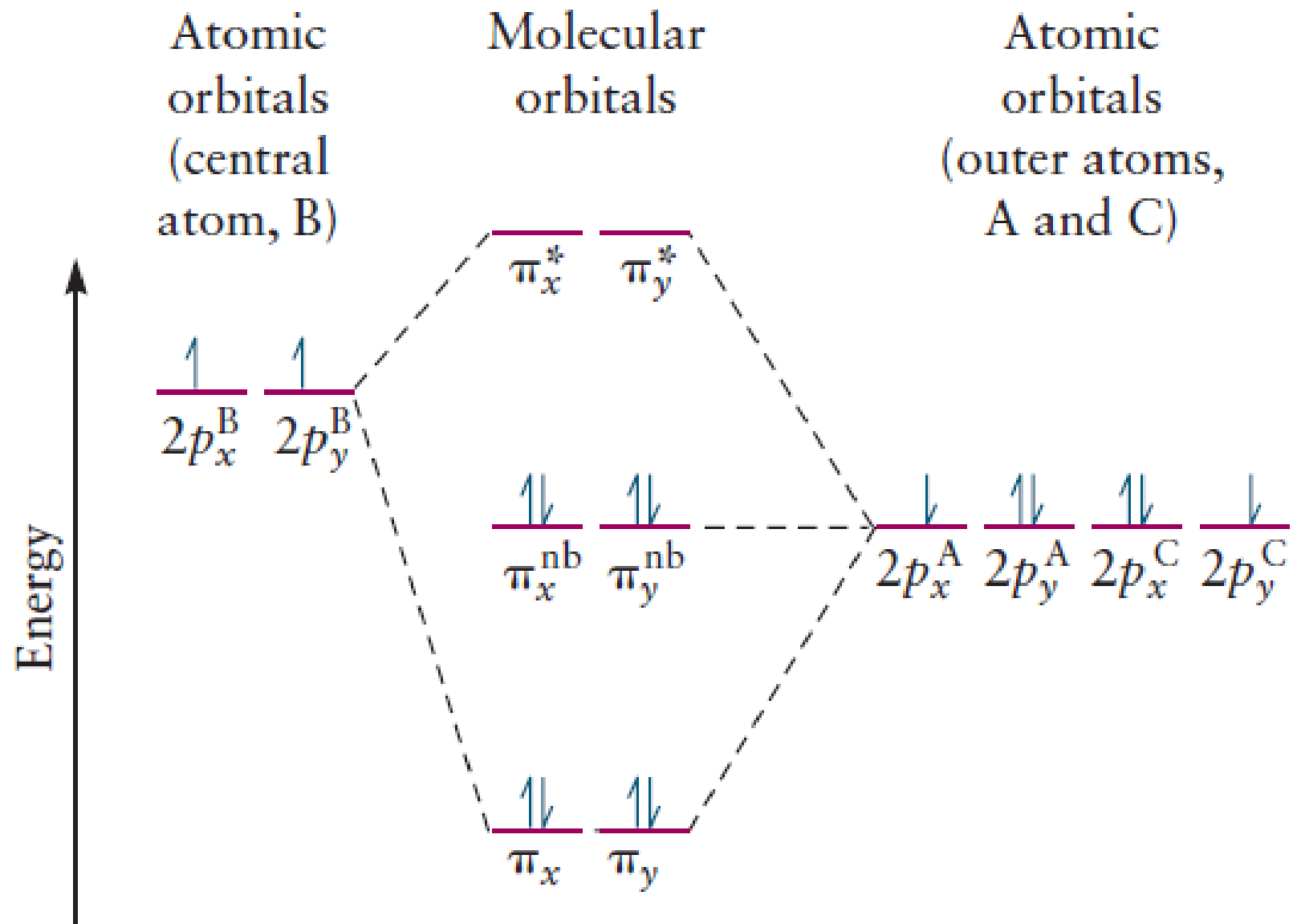


(b) π_x^{nb} nonbonding

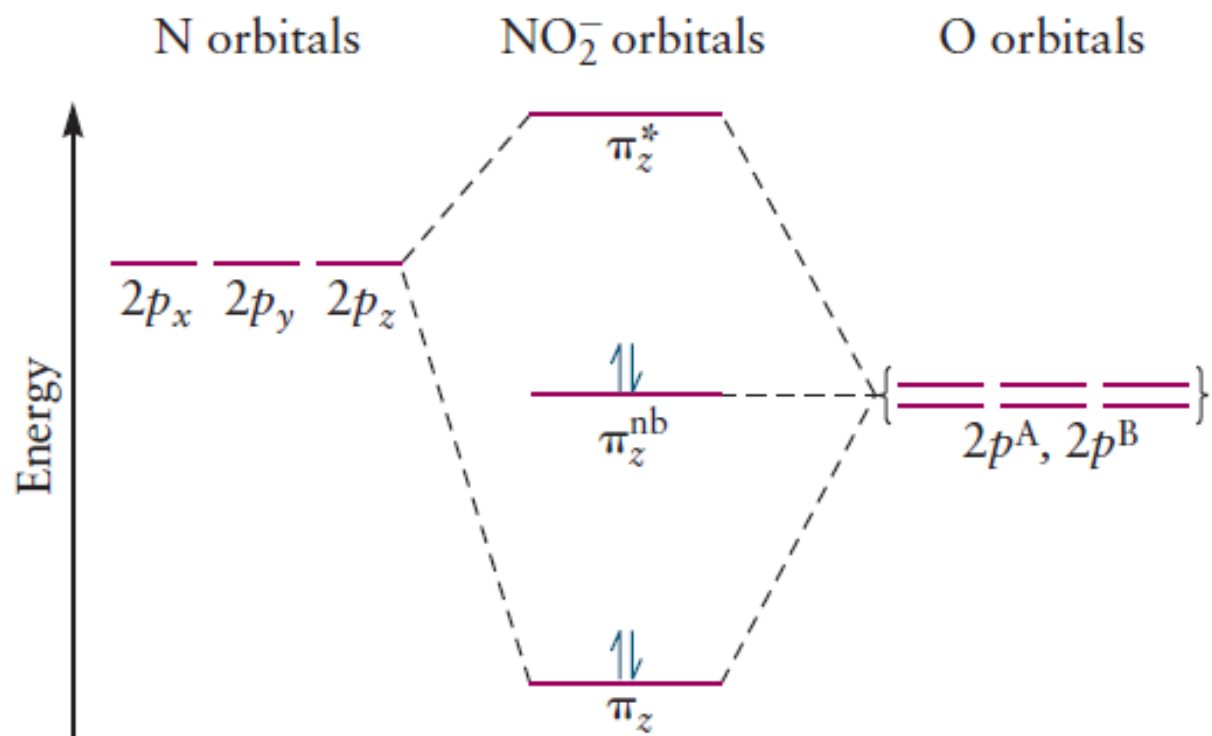


(a) π_x bonding

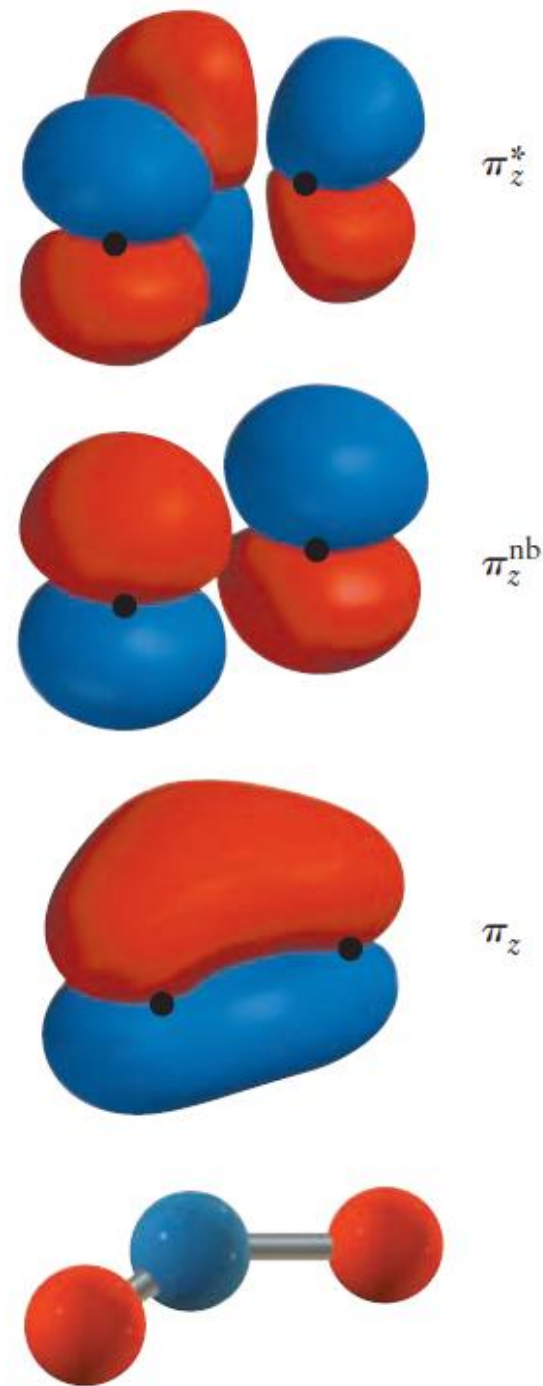
Linear Triatomic Nonhydrides



Nonlinear Triatomic Molecules



(b)



Major points of today's lecture

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Summary and Comparison of the LCAO and Valence Bond Methods

LCAO: constructs **MOs** that are **delocalized** over the entire molecule (**linear combinations of the AOs**)

$$\text{H}_2 \quad \sigma_{g1s} = C_g(R_{AB})[\varphi_{1s}^A + \varphi_{1s}^B] \quad \sigma_{g1s} = [1s^A + 1s^B]$$

$$\psi_{\text{MO}}^{\text{el}} = \sigma_{g1s}(1)\sigma_{g1s}(2) = [1s^A(1) + 1s^B(1)][1s^A(2) + 1s^B(2)]$$

$$\psi_{\text{MO}}^{\text{el}} = [1s^A(1)1s^B(2) + 1s^A(2)1s^B(1)] + [1s^A(1)1s^A(2) + 1s^B(1)1s^B(2)]$$

VB: a quantum mechanical description of the **localized** chemical bond between two atoms

$$\text{H}_2 \quad \psi_{\text{VB}}^{\text{el}}(r_{1A}, r_{2B}) = c_1\varphi_A(r_{1A})\varphi_B(r_{2B}) + c_2\varphi_A(r_{2A})\varphi_B(r_{1B})$$

$$\psi_{\text{VB}}^{\text{el}} = 1s^A(1)1s^B(2) + 1s^A(2)1s^B(1)$$

It assumes a good approximation to the molecular electronic wave function for H_2 is the product of an H(1s) orbital centered on atom A, occupied by electron 1, and another H(1s) orbital centered on atom B, occupied by electron 2.

Summary and Comparison of the LCAO and Valence Bond Methods

LCAO (H_2)

$$\psi_{MO}^{el} = [1s^A(1)1s^B(2) + 1s^A(2)1s^B(1)] + [1s^A(1)1s^A(2) + 1s^B(1)1s^B(2)]$$

VB (H_2)

$$\psi_{VB}^{el} = 1s^A(1)1s^B(2) + 1s^A(2)1s^B(1)$$

- The 1st term in ψ_{MO}^{el} is identical to ψ_{VB}^{el} .
- The 2nd term may be labeled ψ_{ionic} because it is a mixture of the ionic states $H_A^-H_B^+$ and $H_A^+H_B^-$.
- Both methods can be refined to provide more accurate wave functions for molecules and solids.
- Both methods have been improved significantly.

one approach for improving ψ_{VB}

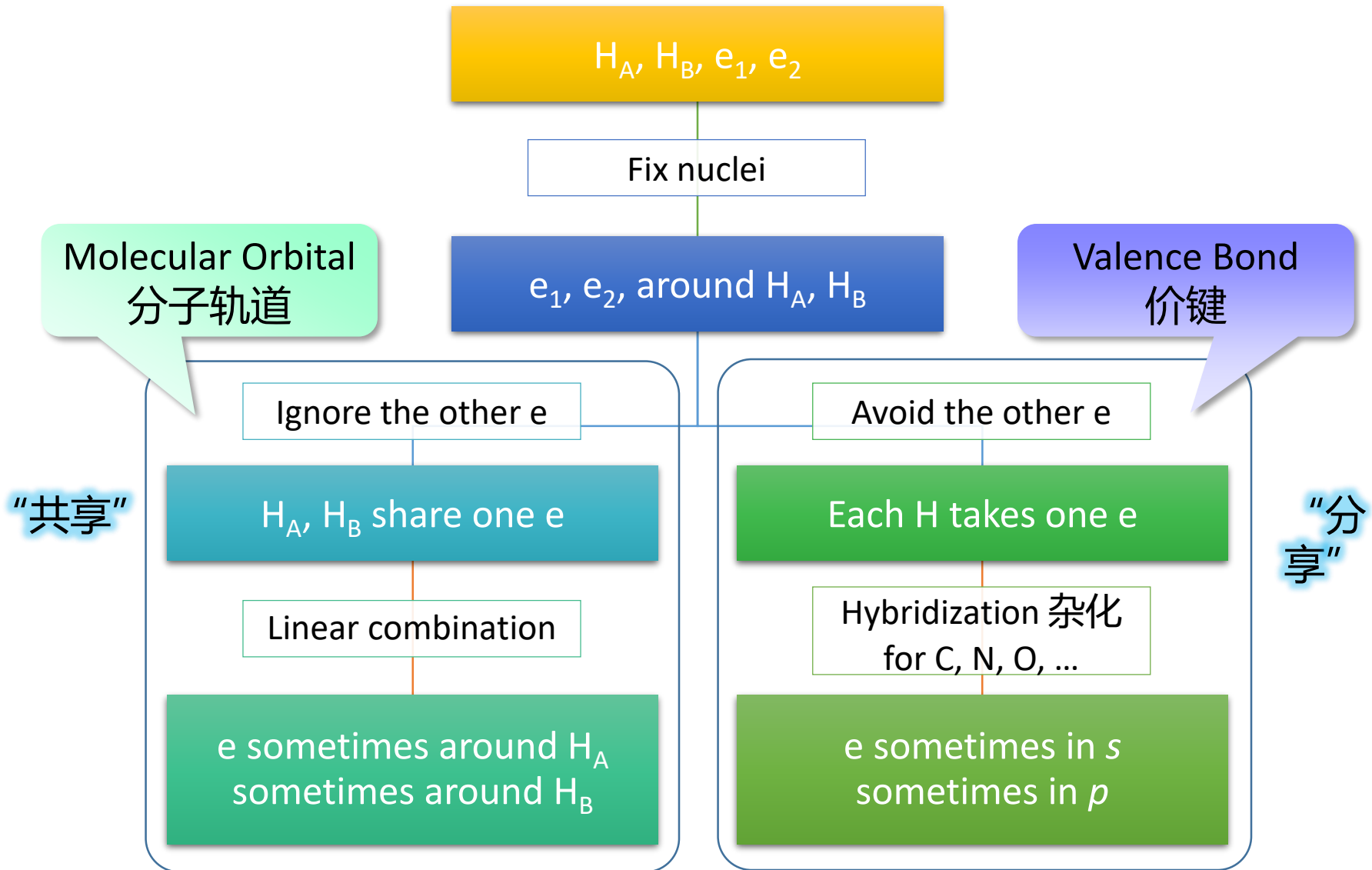
$$\psi_{improved} = \psi_{VB} + \lambda\psi_{ionic}$$

The accuracy of the simple VB wave function can be improved by adding in some ionic character.

Summary and Comparison of the LCAO and Valence Bond Methods

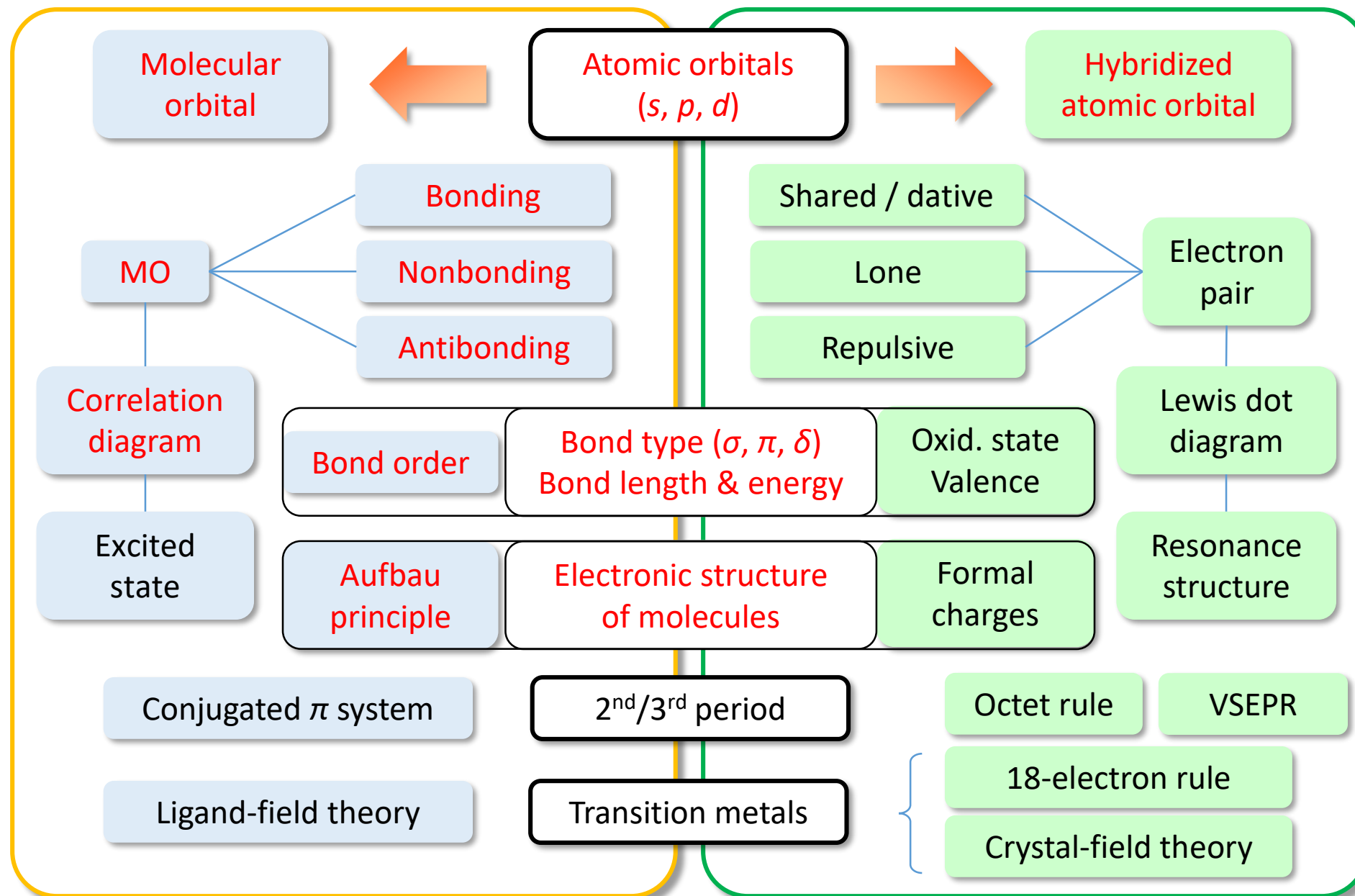
- **LCAO:** describe **the electronic states of molecules** in contexts that require knowledge of energy levels. (Examples include molecular spectroscopy, photochemistry, and phenomena that involve ionization).
- **VB:** describe **molecular structure**, especially in pictorial ways.
- **Localized VB** σ bonds describe the network holding the molecule together
- **De-localized LCAO** π bonds describe the spread of electron density over the molecule.

H₂: Levels of Approximation



LACO (MO)

Valence Bond (VB) / Hybridization



Homework: read “principle of modern chemistry”, chapter 6